

UNIT-1

PC SOFTWARE:-

- ❑ PC software" refers to computer programs or applications that are designed to run on personal computers (PCs).

These software applications are written to perform specific tasks, functions, or operations on a personal computer, enhancing its capabilities and providing users with various features and tools.

PC software can encompass a wide range of applications, from productivity tools to entertainment software.

PC software can be broadly categorized into two main types:

1. System software

2. Application software

- **System Software:** System software includes essential programs that manage and control the basic functions of a computer. This type of software is necessary for the computer to operate and interact with its hardware components. Examples of system software include:
 - **Operating Systems (OS):** The core software that manages hardware resources and provides a user interface. Examples include Windows, macOS, and Linux.
 - **Device Drivers:** Software that allows the operating system to communicate with hardware devices like printers, graphics cards, and peripherals.
 - **Utilities:** Tools for managing and optimizing system performance, disk storage, security, and other system-related tasks.
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- **Device Drivers:** Software that allows the operating system to communicate with hardware devices like printers, graphics cards, and peripherals.
- **Utilities:** Tools for managing and optimizing system performance, disk storage, security, and other system-related tasks.
- **Application Software:** Application software consists of programs designed to perform specific tasks or applications for users. These programs are often what users interact with directly to complete various tasks. Examples of application software include:

Word Processors: Software for creating and editing documents, such as Microsoft Word or Google Docs.

Spreadsheets: Programs like Microsoft Excel or Google Sheets used for managing and analyzing data.

Graphics and Design Software: Applications for creating images, graphics, and designs, such as Adobe Photoshop or CorelDRAW.

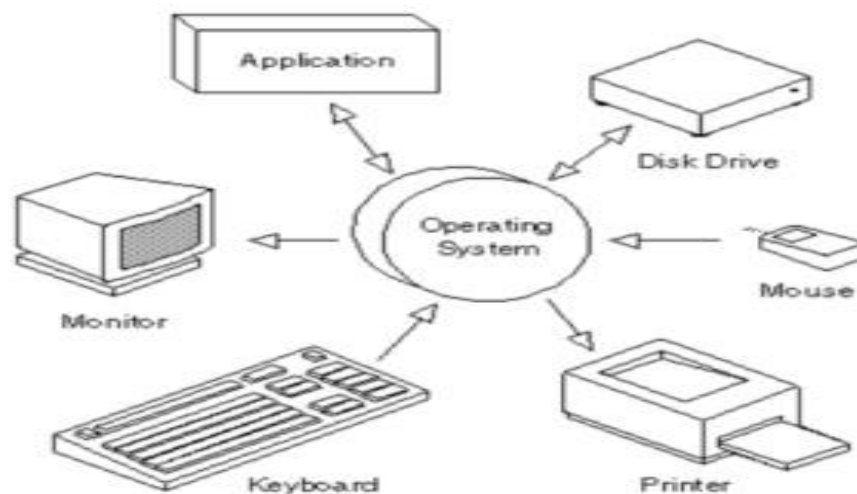
Web Browsers: Software like Google Chrome, Mozilla Firefox, or Microsoft Edge for browsing the internet.

Media Players: Software for playing audio and video files, such as VLC Media Player or Windows Media Player.

OPERATING SYSTEM

- An Operating System (OS) is software that acts as an intermediary between a computer and its user, providing an interface for interaction. It falls under the category of system software.
- The Operating System is a collection of instructions stored on storage devices. It also encompasses a collection of programs that manage a computer's resources and operations.
- The OS is the first program to load when a computer starts. It's often referred to as the "program of programs."
- The OS manages all of the computer's operations.

- The function of an operating system (OS) is to execute other programs and applications, serving as a bridge between the computer's hardware and software.
- A computer is rendered useless without an OS.
- In a multitask operating system, numerous programs can run simultaneously, and the operating system determines when and for how long each program will run.
- **Operating systems can be categorized into two types:**
 - Character User Interface (CUI)
 - Graphical User Interface (GUI)
- **1. Character User Interface (CUI)** – CUI is not very user-friendly, requiring users to type commands to operate the OS. For instance, DOS is a CUI operating system.
- **2. Graphical User Interface (GUI)** – GUI operating systems are user-friendly, eliminating the need for command typing. Instead, users can interact with the OS by clicking on programs using a mouse. Windows is an example of a GUI operating system.



CHARACTERISTICS OF OPERATING SYSTEM

- ❑ **Memory Management:-**In an operating system (OS), memory management involves the management of computer memory, and it keeps track of the complete information about primary memory. It monitors and determines which parts of memory are being used by which programs. Whenever a program requests memory, the operating system allocates it.
- ❑ **Processor Management:-** It allocates the program to the processor (CPU) and, when a program no longer requires the CPU's resources, it deallocates it as well.
- ❑ **Device Management :** The operating system (OS) maintains information about all devices, and this role is also known as I/O (Input/Output) controller. Additionally, the OS makes decisions about which program is given access to which device, when it is given access, and for how long it is granted access.
- ❑ **Security:** It protects any program or data from unauthorized access and uses techniques such as passwords and other security measures.
- ❑ **Booting the Computer:** The process of starting or restarting the computer is known as booting. If the computer is switched off completely and if turned on then it is called cold booting. Warm booting is a process of using the operating system to restart the computer.
- ❑ **Reliability:** It is very reliable because it can detect any viruses and harmful codes.
- ❑ **Cost:** The cost of the operating system is determined based on its features. For example, Windows costs around \$100, while DOS and UNIX operating systems are free.
- ❑ **Ease of use:** It can be easily used because it has a graphical user interface (GUI).
- ❑ **Security:** It protects any program or data from unauthorized access and uses techniques such as passwords and other security measures.

ADVANTAGE OF OPERATING SYSTEM:-

1. **Access the computer:-** with the help of operating system we can access the computer because without operating system our computer is like empty box.
2. **Ease of use:** It can be easily used because it has a graphical user interface (GUI). New users can operate a computer easily through this interface.

3. **Easy updates:** It can be updated easily. Multiple software are present, issuing regular updates to fix bugs and launching new features. In the same way, OS issues updates at regular intervals, fix bugs and introduce new features.

4. **Security:** It is secure, for example, in Windows, there is Windows Defender that can detect harmful files and remove them.

5. **Software installation:** We can install application software in our computer and run any games or software through it.

6. **Open source:** Some operating systems, like Linux, are open source, and we can run them on our computer for free.

7. **Multitasking:-** OS has ability to execute multiple tasks at a time. we can run multiple application in our computer at one time with the help of OS.

DISADVANTAGE OF OPEARTING SYSTEM

1. **System Failure:** OS is considered to be the heart of the computer system. Without the presence of the Operating System, computers will not operate. With the crash OS, no functions, tasks, and commands will be executed. Several reasons are provided for the system failure, and they are:

- Malware and viruses corrupt the computer system
- Improper, bug-filled devices

2. **Operating System Costs:** Some operating systems are free, but others can be expensive. For example, the price of Windows is typically around ₹5,000 to ₹10,000.

3. **Linux Usability:** Running Linux can be a bit challenging compared to Windows.

4. **Hardware Compatibility:** It sometimes doesn't support certain hardware.

5. **Mac OS Security:** Mac OS is more susceptible to viruses.

WINDOWS OPERATING SYSTEM:-

The Windows operating system was developed by **Microsoft Corporation**, an **American Multinational technology company** founded by **Bill Gates** and **Paul Allen**.

- Windows is a **graphical operating system** developed by Microsoft.
- **Windows Operating System** is a type of operating system based on **Graphical User Interface (GUI)**. It is called a graphical user interface because it create an interface (communication screen) between user and machine by using graphical elements like icons, pictures, colors, etc.
- It allows users to view and store files, run the software, play games, watch videos, and provides a way to connect to the internet.
- It was released for both home computing and professional works.
- Microsoft introduced the first version as 1.0
- It was released for both home computing and professional functions of Windows on **10 November 1983**.
- Later, it was released on many versions of Windows as well as the current version, Windows 10.

COMPONENT OF WINDOWS:-

1. Desktop: Desktop is the on screen work area on which window appears. A Desktop is main opening screen when operating system is started on a computer.

- **Taskbar:** The Taskbar appears on the bottom of the screen. On the bottom left corner, there is the area for quick launch icons then the programs running, and lastly the notifications area.
- **Quick launch icons:** It contains icons that are used to access a particular program quickly.
- **Running Applications:** This area shows the buttons of the programs that are currently open.
- **Notification area:** It can be about automatic updates, the antivirus program etc. The time also appears here. To open them simply double-click on the icon.
- **MY Computer:** This is a standard icon and is very useful and provides access to the resources on the computer. It allows to control and view disk drives.
 - Double click on Computer to view its content.
 - Close the windows by clicking on the X in the upper right hand corner of the screen.
- **Documents:** Documents is a folder that provides a convenient place to store the documents, graphics and other files for quick access.
- **Recycle Bin:** When one deletes a file, windows sends it to the recycle bin instead of deleting it. It allows to restore the file in case it is deleted by mistake.

2. Window: A Window is an area on the desktop within which all windows based program run.

- **An Application Window:** An Application window is a running application such as MS Word, MS Excel, Paint, Photoshop etc. These applications can be opened and run simultaneously, but there is only one active window at a time.

- **The various components of an application window are:**

- **Title Bar:** Title Bar displays the document name, program name and standard Windows control buttons. Title bar is located at the top of the screen. **On the right side of the Title bar are three buttons:**
 - The Minimize button makes the window disappear from the screen without closing the program.
 - The Reduce/Maximize button makes the window smaller or takes it back to full size.
 - The Close button closes the document or program.
- **Menu Bar:** It is located below the title bar. It consists of various menu options. Each menu option is associated with different types of pull down menu lists. On clicking any menu option a pull down menu is displayed.
- **Toolbars:** Toolbars generally appear below the menu bar. One can drag them and display them along any of the window borders. The icons on the toolbar are used to send commands to the program.
- **Work Space:** It is the rectangular area in a window below the bars where the application or document opens and displays the contents of the currently active file.
- **Scroll Bars:** Scroll Bars are used to move the document. There are two scroll bars. One is at right side of the screen called vertical scroll bar. The second is at the bottom of the screen called horizontal scroll bar. The user can scroll the document by clicking the scroll arrows at either end or by dragging scroll button.
- **Status Bar:** It is located at the bottom of the window. This bar displays information about the document on which the user is currently working.

ADVANTAGES OF WINDOWS OPERATING SYSTEM:

- **User-Friendly Interface:** Windows provides a familiar and user-friendly graphical user interface (GUI) that is easy for most people to navigate.

- **Wide Software Compatibility:** It supports a vast library of software applications, including productivity tools, games, and business software.
- **Hardware Compatibility:** Windows OS is compatible with a wide range of hardware devices, making it suitable for various computer configurations.
- **Security Features:** Windows includes built-in security features, such as Windows Defender antivirus, Windows Firewall, and regular security updates.
- **Multitasking:** Windows supports multitasking, allowing users to run multiple applications simultaneously.
- **Updates and Support:** Microsoft provides regular updates, bug fixes, and technical support for its operating systems.

DISADVANTAGES OF WINDOWS OPERATING SYSTEM:

- **Cost:** Windows OS typically requires a licensing fee, which can add to the cost of setting up and maintaining a computer system.
- **Virus Attack :-** as compare to other operating system, in windows operating there are more chance for attacking . In windows operating system users have to depend on antivirus.
- **Compatibility Issues:** Older software or hardware devices may not be compatible with newer versions of Windows, requiring workarounds or replacements.
- **Rebooting:** - if you load many software at same time then your system slower down and hang up.

ICONS:-

- Icon is small graphical pictures on screen.
- Icons on a computer refer to the small graphical representations or symbols used to represent files, folders, programs, and other elements on the computer's graphical user interface (GUI).
- Icons serve as visual shortcuts, making it easier for users to identify and interact with various items on their computer.
- To activate an icon double click on it with the left mouse button, this will activate the icon and either start a program or open a file/folder.

- The icons on the desktop can be renamed by right clicking on them and selecting rename similarly they can be deleted by right clicking and selecting delete.

TYPES OF ICONS:-

1. Desktop Icons: These are icons that appear on your computer's desktop screen. They often represent files, folders, applications, or shortcuts to programs or documents. Common desktop icons include the Recycle Bin, My Computer (or This PC), and any files or folders you place on your desktop for quick access.

2. System Icons:- those icons which is loaded automatically when you load operating system is called system icons.eg. My computer, Recycle bin, control panel etc.

3. Application Icons: These icons represent software applications installed on your computer. When you double-click an application icon, you can open your file and perform your task. It is also known as program icons. Eg:- Ms word , excel, tally etc

4. Document icons:- The created any program is called document icons. Eg:- word, excel, power point

5.folder icons:- folder is a place where we can store any types of files as like. Eg:- photo, audio, any document files.

- the default color of folder is yellow

6 .Disk Drive Icons: My Computer icon contains the different disk drives like Pen Drive, Hard Disk etc.



Various Operations performed with icons:

- **Opening a file or program:** When you double-click or open an icon, the associated file or program will be opened.
- **Rename:** You can quickly rename any highlighted windows icon by pressing the F2 key on the keyboard and typing in a new name for the icon.
- **Shortcuts:** A Shortcut Icon will open a program that someone has added to the Desktop. The following icon has a little black arrow in it.
- **Changing an Icon:** The icons can be changed as per requirement by right-clicking on the icon that you want to change. Click on Properties. Select the Customize tab. Click on Change Icon button. Select an icon. Click on Apply.

WINDOWS EXPLORER

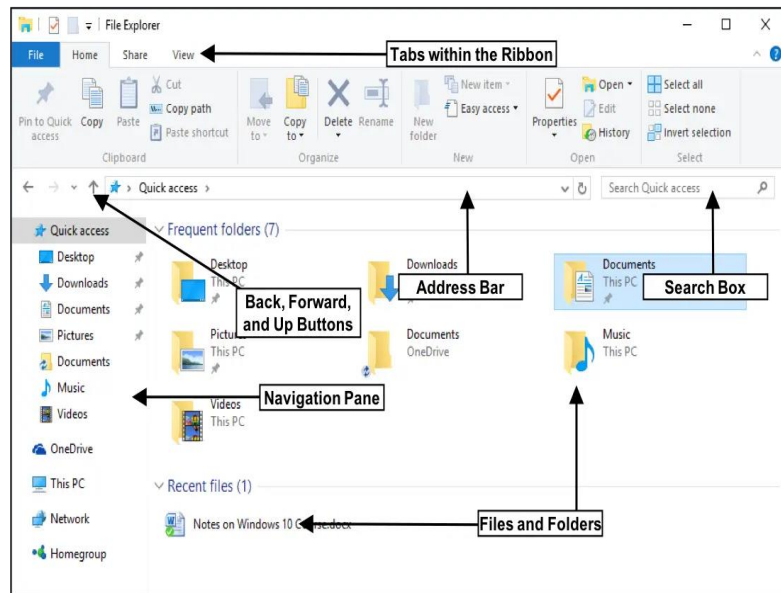
- Windows Explorer is the former name of the file management application and user interface component in Microsoft Windows operating systems
- IN windows old version is known as window explorer but in window 10 is known as file explorer.
- Windows Explorer or File Explorer is a fundamental tool for navigating and managing files and folders on a Windows computer. It provides graphical interface for various file-related tasks.
- File Explorer is an essential tool for managing files and folders in the Windows operating system, and it plays a crucial role in everyday computing tasks.
- **These are following function of windows explorer:- ▼**

- **To start File Explorer in Windows 10**, click the “File Explorer” app within the “Windows System” folder in the “All apps” listing of the Start menu or Start screen.

- **Alternatively**, click the “File Explorer” button in the Taskbar at the bottom of the desktop.

- The **Navigation Pane** at the left side of the File Explorer window lets you select folders, drives, and also navigate your network’s computers, folders, and files.

- The **Back, Forward, and Up Buttons** let you navigate through the screens you browse within File Explorer.



- **To return to the last page viewed**, click the **Back** button.
- Click the **Forward** button to move forward again after clicking **Back**.
- **To move up a level in the folder hierarchy shown within File Explorer**, click the **Up** button.
- After selecting an item from the **Files and Folders listing** in File Explorer, all its commands appear on the **Tabs within the Ribbon** at the top of File Explorer. Clicking the tabs shows the functions available in each group.
- The **Address Bar** shows the currently selected computer or network location.
- **File Preview:** Users can preview the contents of files (such as text, images, or documents) without opening them fully. This feature is useful for quickly checking the contents of files before opening them.

FILES:

- A Collection of information stored into one group on a disk is called file.
- Infact, almost all the information stored in a computer is stored in the form of files.
- Thus files are collections of information.

- There are many types of files, including operating system files, program files and one's own created files.

FOLDERS:

- A Folder is a location in which one can store files and other folders.
- A folder is similar to a drawer in which one keeps all the files.
- To keep the files organized, one must create folders and store the files in the folder.

➤ Steps to Create a Folder:

- Open My Computer. Double click the disk drive or folder where to create a new folder.
- On the File menu, point to New and then click Folder.
- A new folder icon appears with its name highlighted. The default name is New Folder.
- Type a new more descriptive name and press enter.

➤ Copying and Moving File and Folder:

- Select the file or folder that one want to move or copy.
- Click Edit option
- To move the file, click Cut
- To copy the file, click Copy
- Open the folder where one wants to place the file and click. Use Paste option to move or copy the file.

ACTIVATING WINDOWS

1. Steps to activate Windows 10 with a product key

STEP 1

For installation of Windows 10, enter your product licence key.

STEP 2

Press the Windows key, go to Settings > Update and Security > Activation.

STEP 3

Press the Change Product key.

STEP 4

Enter your product key into pop-up box and press Next.

STEP 5

Press Activate.

WINDOWS ACCESSORIES:

- Windows operating system has its own programs that one can use to perform simple tasks such as writing letters, paint pictures, and perform calculations.
- These program used to perform small task and get load automatically when we load windows operating system in our computers.
- Each of these programs is robust and offers some fairly advanced features that can be used to handle various tasks.

Options available with Windows Accessories:

- **Calculator:** Calculator is used for performing calculations. The calculator is available in two views:
 - Standard View
 - Scientific View

To Start Calculator perform the following steps:

- Click Start then All Programs, point to Accessories and select Calculator.
 - Calculator window opens up on the screen.
 - With a click on the view tab of calculator window of step (b), one can select the other type of calculator.
- **Notepad:** Notepad is a simple text editor. Text editor means a program in which one can view and edit text. It is used to create or edit text files which do not require formatting. These text files must be smaller than 64 Kilobytes. The default extension of all the notepad files is **.txt**
 - **To Start the notepad, the steps are:**
 - Click Start then All Programs, point to Accessories and select Notepad. The notepad screen will appear.
- **WordPad:** WordPad like Notepad is a text editor but it is more elaborate than Notepad. Wordpad has more features than Notepad. One can format documents in WordPad with various fonts and WordPad files is **.rtf**. It is used to create formatted documents that can be opened by any word processing program.

- **To Start the WordPad, the steps are:**
 - Click Start then All Programs, point to Accessories and select WordPad. The WordPad screen will appear.
- **Paint:** Paint is used to create and edit pictures, drawings and graphic images. One can paste a Paint picture into another document or use it as desktop background or even create scanned photos. One can become a good computer artist with the help of Paint. One can make attractive logos, draw diagrams, use scanned photographs, colour pictures and even create new desktop wallpaper designs.
 - **To Start the Paint, the steps are:**
 - Click Start then All Programs, point to Accessories and select Paint. The Paint window will appear.
 - **Features of Paintbrush:**
 - It is very simple and accessible user interface.
 - Paintbrush is a simple little image editing application.
 - It allows selection of drawing tools.
 - It provides reasonable range of options for adding text.
 - It is a straight forward user interface where users find no trouble in finding their way to artistic creations.

